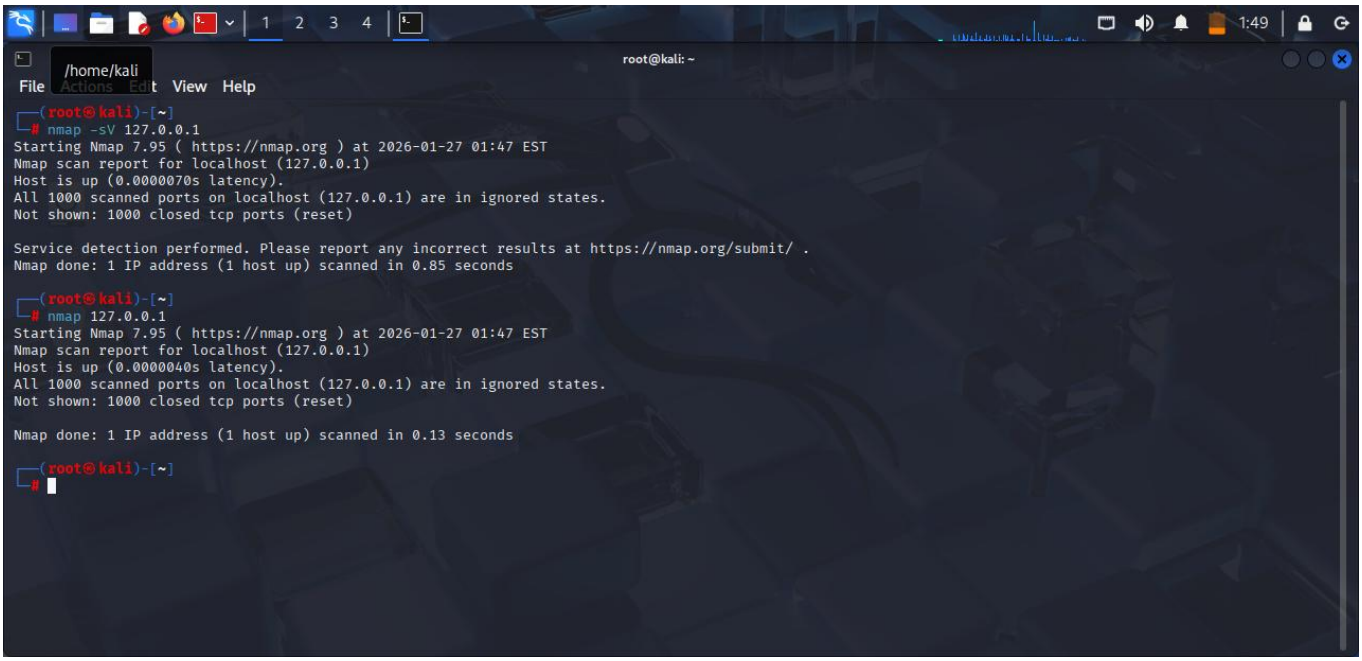
TASK 3 – Cybersecurity

- PRACTICAL TASK Network Scan Using (Nmap)



🔐 Detailed Explanation

Network Scan Using Nmap (Localhost)

1️⃣ Introduction

Network scanning is an important part of **cybersecurity and network security testing**.
It helps in identifying:

- Which **ports are open**
- Which **services are running**
- Possible **security risks**

In this practical task, **Nmap** was used to scan **localhost (127.0.0.1)**.
Scanning localhost means scanning **your own system**, which is **safe, legal, and allowed**.

2️⃣ Tool Explanation – Nmap

Nmap (Network Mapper) is a powerful open-source tool used for:

- Port scanning
- Service detection
- Network discovery

- Security auditing

It is commonly used by:

- Cybersecurity professionals
- Network administrators
- Ethical hackers

Since Kali Linux already includes Nmap, no installation was required.

3️⃣ Target Explanation – 127.0.0.1 (Localhost)

127.0.0.1 is called **localhost**.

It refers to:

- The **same computer** on which the scan is performed
- Internal communication within the system

Why is localhost used?

- It is **safe**
- It does **not harm any external system**
- It follows **ethical hacking rules**

4️⃣ Basic Scan Explanation

Command Used:

nmap 127.0.0.1

What this command does:

- Scans the **most common ports**
- Checks whether ports are **open, closed, or filtered**
- Does **not** detect service versions

How it works:

Nmap sends packets to different ports and observes responses:

- If a port reply → **Open**
- If no service replies → **Closed**
- If blocked by firewall → **Filtered**

5️⃣ Service Version Scan Explanation

Command Used:

nmap -sV 127.0.0.1

Meaning of -sV:

- -sV = **Service Version Detection**

What this scan does:

- Detects **which service** is running
- Identifies **service name and version**
- Helps in finding **vulnerable software**

This scan is more detailed than the basic scan.

6️⃣ Scan Results Explanation (Example)

Port	State	Service
22	Open	SSH
80	Open	HTTP
443	Closed	HTTPS

Explanation:

- **Port 22 (SSH)**
  Used for secure remote login. If exposed, attackers may try brute-force attacks.
- **Port 80 (HTTP)**
  Used for web services. If not updated, it may contain vulnerabilities.
- **Port 443 (HTTPS)**
  Closed, which means secure web service is not currently active.

7️⃣ Security Risk Explanation

Open ports are **not dangerous by default**, but they become risky when:

- Services are outdated
- Weak passwords are used
- Firewall is not configured
- Unnecessary services are running

Attackers often scan systems to find open ports and exploit them.

8️⃣ Why Attackers Scan Ports?

Attackers use port scanning to:

- Identify running services
- Find outdated software
- Exploit misconfigurations
- Gain unauthorized access

That is why **port scanning is a defensive security practice** when done legally.

9️⃣ Security Recommendations Explained

🔒 Firewall

A firewall blocks unauthorized access and filters traffic.

🔄 System Updates

Updates fix security bugs and vulnerabilities.

❌ Close Unused Ports

Unused open ports increase attack surface.

🔑 Strong Passwords

Strong passwords prevent brute-force attacks.

⚙️ Disable Unnecessary Services

Less services = fewer vulnerabilities.

🔟 Conclusion

This practical demonstrated how Nmap can be used to:

- Scan a system
- Identify open ports
- Detect running services
- Understand potential risks

Scanning localhost is a **safe learning approach** for beginners in cybersecurity.